

Computer Assignment 1

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```
library(RCurl)

# Loading data
set.seed(031026)
länk <- "https://raw.githubusercontent.com/mskoldSU/MT5003_HT17/master/Projekt/proj_data.csv"
data_individ <- read.csv(text = getURL(länk))
idx <- sample(1:nrow(data_individ), 1000)
data_individ <- data_individ[idx, ]
save(data_individ, file = "proj_data.Rdata")
head(data_individ)

##      Resultat    Kon Alder  Utbildare
## 10314     FALSE    Man   32  Privatist
## 8984      FALSE Kvinna  28 Trafikskola
## 3682      FALSE    Man   19  Privatist
## 5977       TRUE    Man   21  Privatist
## 2299      FALSE    Man   19  Privatist
## 8636      FALSE    Man   27  Privatist

load("proj_data.Rdata")

y <- matrix(data_individ$Resultat, ncol = 1)
y <- as.numeric(y)
X <- model.matrix(Resultat ~ Kon + Alder + Utbildare,
                  data = data_individ)
head(X)

##      (Intercept) KonMan Alder UtbildareTrafikskola
## 10314           1      1   32                0
## 8984            1      0   28                1
## 3682            1      1   19                0
## 5977            1      1   21                0
## 2299            1      1   19                0
## 8636            1      1   27                0

head(data_individ[, -1])

##      Kon Alder  Utbildare
## 10314    Man   32  Privatist
## 8984  Kvinna  28 Trafikskola
## 3682    Man   19  Privatist
## 5977    Man   21  Privatist
## 2299    Man   19  Privatist
## 8636    Man   27  Privatist
```

```

## Functions
## test1
# -----a-----
L <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  prod(dbinom(y, size=1, prob=p))
}

l <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  # p <- plogis(z)
  p <- 1/(1+exp(-z))
  sum(dbinom(y, size=1, prob=p, log=TRUE))
}#using 'dbinom' in both L and l

S <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  t(X) %*% (y-p)
}

I <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  D <- diag(as.vector(p*(1-p)))
  t(X) %*% D %*% X
}

# -----b-----
NR <- function(theta0, niter, y, X){
  for (i in 1:niter) {
    score <- S(theta0, y, X)
    fisher <- I(theta0, y, X)
    theta0 <- theta0 + solve(fisher, score)
  }
  return(theta0)
}

# Using R-package directly
modell1 <- glm(Resultat ~ Kon + Alder + Utbildare,
              data = data_individ,
              family = "binomial")
summary(modell1)

##
## Call:
## glm(formula = Resultat ~ Kon + Alder + Utbildare, family = "binomial",

```

```
##      data = data_individ)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.065131   0.253358  -0.257  0.79712
## KonMan         0.100567   0.137049   0.734  0.46307
## Alder          -0.025122   0.008671  -2.897  0.00376 **
## UtbildareTrafikskola 0.947498   0.158405   5.982 2.21e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1344.4  on 999  degrees of freedom
## Residual deviance: 1291.4  on 996  degrees of freedom
## AIC: 1299.4
##
## Number of Fisher Scoring iterations: 4

# coefficient of glm method
theta_mo <- coef(model1)
theta_mo <- as.numeric(theta_mo)
```

Test 2

```
# Using NR method
theta0 = c(0,0,0,0)
max_iter <- 100
for (i in 1:max_iter){
  theta0 <- NR(theta0, niter=1, y, X)
  theta_hat <- theta0
  if (all(round(theta_hat, 2) == round(theta_mo, 2))) {
    cat("niter =", i, "reaches two-digits accuracy")
    break
  }
}

## niter = 2 reaches two-digits accuracy
```

Test 3

```
# Calculate ML-estimates standard errors
ML_Std_Error <- function(theta_hat, y, X){
  fisher <- I(theta_hat, y, X)
  fisher_inv <- solve(fisher)
  ml_std_error <- sqrt(diag(fisher_inv))
}

ML_esti_se <- ML_Std_Error(theta_hat, y, X)
print(ML_esti_se)
```

##	(Intercept)	KonMan	Alder
##	0.253340181	0.137041498	0.008669758
##	UtbildareTrafikskola		
##	0.158401659		

Conclusion: The comparison between the standard errors of the ML estimates and those from the GLM model shows agreement to at least four-digit accuracy, indicating that R is using this method.

Test 4

```
# Using parametric Boothstrap
# Set the number of bootstrap samples
n <- 1000

# Store bootstrap parameter estimates
boot_esti <- matrix(NA, nrow = n, ncol = length(theta_hat))

# My age: 22; sex: woman
# Store bootstrap probabilities for 22-year-old privately educated woman
n
boot_probs <- numeric(n)

# Parametric bootstrap
for (b in 1:n) {
  # Generate new response values using estimated probabilities
  prob_esti <- 1 / (1 + exp(-X %*% theta_hat))
  y_new <- rbinom(n = length(y), size = 1, prob = prob_esti)

  # Estimate parameters
  theta_boot <- NR(theta0, niter = 1, y_new, X)
  boot_esti[b, ] <- theta_boot

  # Create predictor vector for 22-year-old privately educated woman
  x_new <- c(1, 0, 22, 0)

  # Calculate probability for this individual
  linear_pred <- sum(x_new * theta_boot)
  prob <- 1 / (1 + exp(-linear_pred))
  boot_probs[b] <- prob
}

# Calculate bootstrap standard errors
boot_se <- apply(boot_esti, 2, sd)
print(boot_se)

## [1] 0.260405592 0.140081991 0.008773274 0.158770335
```

```
# Calculate 95% CI for the probability
ci_prob <- quantile(boot_probs, probs = c(0.025, 0.975))
print(ci_prob)

##      2.5%      97.5%
## 0.2988193 0.4080682
```